

- Q.30 What is the biological effect of Ultrasound.
- Q.31 Write a short note on different scan used in ultrasound.
- Q.32 Explain Larmor frequency and dipole moment.
- Q.33 Explain intensifying screen.
- Q.34 Explain the role of biomedical engineering in radiology services.
- Q.35 Explain properties and applications of ultrasound waves.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain various types of interactions between x-rays and matter with suitable diagram.
- Q.37 Explain construction, types and uses of NMR Coils in MRI with block diagram.
- Q.38 Explain mammography and its application with block diagram.

No. of Printed Pages : 4

183253

Roll No.

5th Sem / Med. Eltx Subject:- Medical Imaging Techniques (MIT)

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is used to convert x-rays into visible lights in radiography screens
- a) Selenium b) Cesium Iodide
- c) Lead d) Sodium chloride
- Q.2 The primary purpose of collimator in x-ray machine is to
- a) Protect the operator
- b) Focus x-ray beam
- c) Reduce noise
- d) Improve patient positioning
- Q.3 The process of converting electrical energy into x-rays occurs in the
- a) Filament b) Cathode
- c) Anode d) Grid
- Q.4 Gantry houses all of the following except
- a) X-ray tube b) Detector
- c) Patient couch d) Collimator

(20)

(4)

183253

(1)

183253

- Q.5 The component responsible for detecting x-ray that pass through the body in CT
- Film
 - Photo-diode array
 - Monitor
 - Ion chamber
- Q.6 The main magnet used in MRI is made up of
- Copper
 - Aluminium
 - Superconducting Magnets
 - Permanent Magnets only
- Q.7 Which is not a part of ultrasound machine
- Magnet
 - Transducer
 - Display
 - Receiver
- Q.8 A-Scan ultrasound used in
- Abdominal scanning
 - Ophthalmology
 - Cardiology
 - Obstetrics
- Q.9 Doppler ultrasound is used for
- Bone structure
 - Lung ventilation
 - Blood flow
 - skin elasticity
- Q.10 Which influences the resolution in ultrasound imaging
- wave speed
 - sound waves frequency
 - patient height
 - battery power

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write wavelength range of x-ray.
- Q.12 Name any one interaction between x-ray & matter.
- Q.13 What does anode do in x-ray tube.
- Q.14 Photoelectric effect.
- Q.15 Thermionic emission.
- Q.16 Name one contrast used in Angiography
- Q.17 Define contrast media.
- Q.18 Expand NMR.
- Q.19 Expand DEXA
- Q.20 What is CT Number.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain electromagnetic spectrum.
- Q.22 Explain difference between coherent & Compton scattering.
- Q.23 Write a note on generation and detection of x-rays.
- Q.24 Write a note on construction and need of beam restrictors and grids.
- Q.25 Explain gantry movement.
- Q.26 Explain Angiography procedures.
- Q.27 Write a note on biological effects of ionizing radiations.
- Q.28 Explain pulse-echo techniques.
- Q.29 Explain principle of ultrasound transducer.